

Week 5 – Lecture 2

Artificial Intelligence Foundations

Unit II: Reasoning Under
Uncertainty

Bayes' Theorem

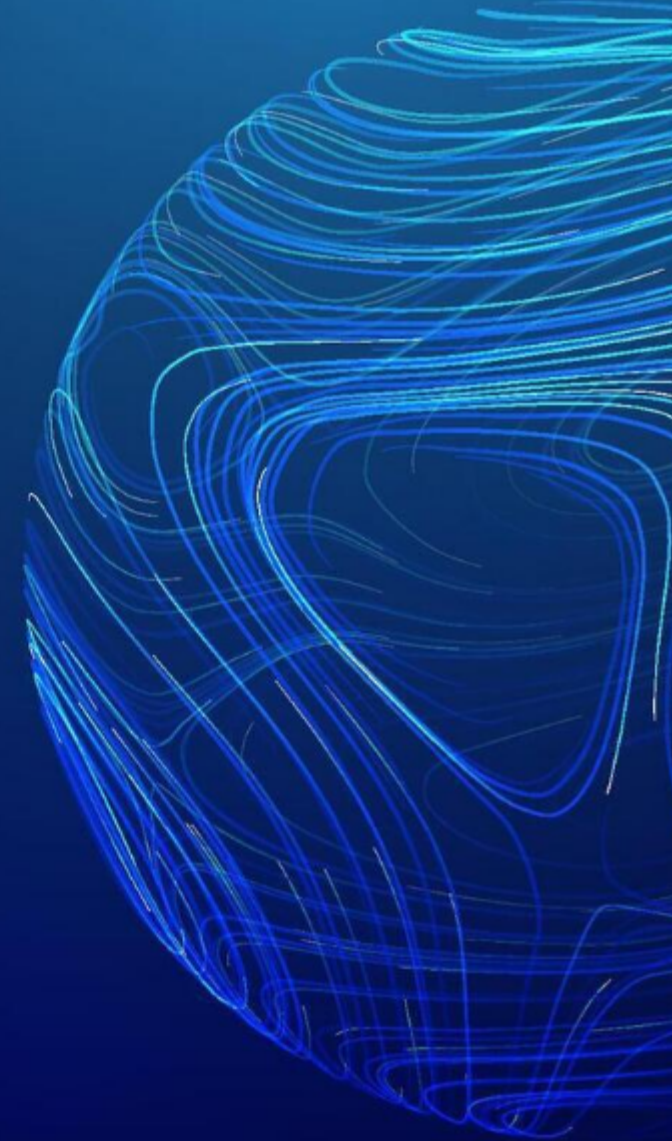
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Motivation: Real-World AI

Can artificial intelligence ever be completely certain?



Medical Diagnosis

Doctors and AI systems rarely have 100% certainty based solely on observing symptoms.



Spam Detection

An email containing the word "free" might be spam, but it could also be a legitimate newsletter.



Weather Prediction

Atmospheric data is noisy. Forecasts are inherently probabilistic, not absolute guarantees.

What is Uncertainty in AI?

In classical logic, statements are either **True (1)** or **False (0)**. However, the real world is filled with ambiguity, noise, and incomplete data.

Therefore, modern AI often works with **probabilities** instead of absolute truth to quantify uncertainty.

Common Uncertainties:

- Will it rain tomorrow?
- Is this email actually spam?
- Does this patient have the disease?
- Will this customer repay their loan?



Why Logic Alone Isn't Enough



Deterministic Logic

Requires strict, absolute rules.

```
IF Symptoms = Fever  
THEN Disease = Flu
```

Fails when data is ambiguous or rules overlap (e.g.,
COVID also causes fever).



Probabilistic Reasoning

Operates in the gray area of the real world.

```
IF Symptoms = Fever  
THEN P(Flu) = 60%, P(COVID) = 30%
```

Allows AI to make the best possible decisions despite
uncertainty.

Probability as a Measure of Belief

In AI, we view probability as our **degree of confidence** in a specific outcome.



0%

50%

100%

(e.g., Rolling a 7 on a standard die)

(e.g., Flipping a fair coin)

(e.g., The sun rising tomorrow)

AI models rarely output exactly 0% or 100%. They output **probabilities** that guide decision-making thresholds.

Interactive Discussion

80%

Chance of Rain

Would you carry an umbrella today? Why?

Even without absolute certainty (100%), humans routinely make rational decisions based on available probability.

This is exactly how Artificial Intelligence handles real-world problems: by updating its confidence and acting on the most likely outcome.

Introducing Bayes' Theorem

The core idea: How should we change our beliefs when we see new data?



Initial Belief

What we know before seeing any new evidence (Prior).



New Evidence

New data, test results, or observations we collect.



Updated Belief

Our revised confidence after considering the evidence (Posterior).

Understanding the Formula

$$P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$$

P(A|B)

Posterior

Updated belief that A is true,
given the new evidence B.

P(B|A)

Likelihood

Probability of seeing the
evidence B, if A is indeed true.

P(A)

Prior

Initial probability of A being true,
before evidence B.

**P(B)
Evidence**

Total probability of observing
evidence B under any condition.

Everyday Analogy: The Detective



Prior Belief

Suspect A had a motive. The detective believes there is a **20%** chance Suspect A is guilty.



New Evidence

Suspect A's fingerprints are found at the crime scene. This is highly unlikely if they were innocent.



Updated Belief

Applying Bayes' logic, the detective's confidence in Suspect A's guilt skyrockets to **95%**.

Medical Diagnosis Example

How doctors (and medical AI) synthesize information:



1. Base Rate (Prior)

Before any testing, the probability a random person has Disease X is very low (e.g., 1%).



2. Medical Test (Evidence)

The patient takes a test that is 95% accurate. The result comes back **Positive**.



3. Diagnosis (Posterior)

We combine the rarity of the disease with the accuracy of the test to find the true probability they are sick.

Numerical Example: Let's Do the Math

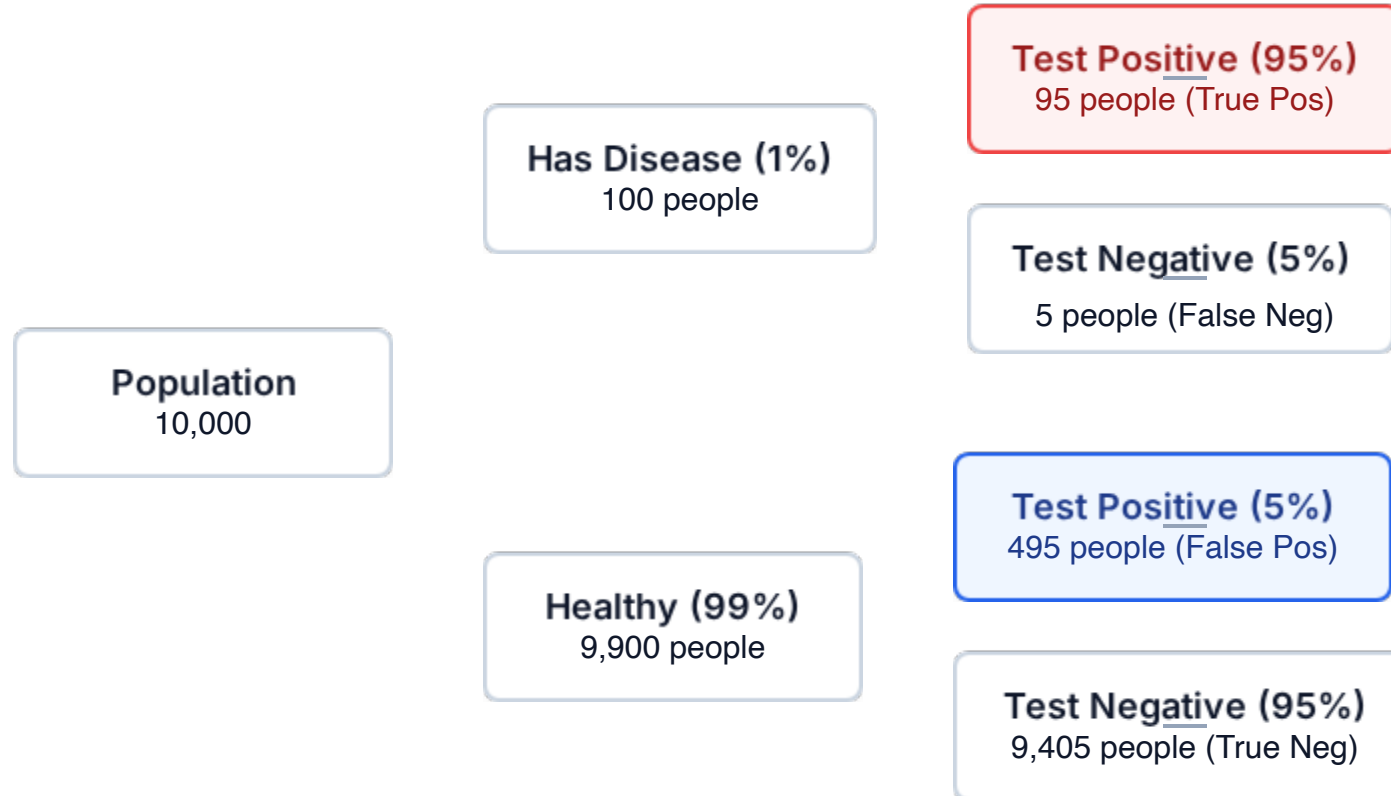
The Scenario:

- **Disease Prevalence:** 1% of the population.
- **Test Accuracy:** 95%. (It correctly identifies 95% of sick people, and 95% of healthy people).
- **Situation:** You test Positive.

Let's look at a population of 10,000 people to make it intuitive.

Status	Test Positive	Test Negative
Disease (100)	95 (True Pos)	5 (False Neg)
Healthy (9,900)	495 (False Pos)	9,405 (True Neg)

Probability Tree Visualization



Analyzing the Results

16%

Actual Probability

Does a positive result mean you are definitely sick?

Even though the test is **95% accurate**, there are 590 total positive tests (95 True + 495 False). Your chance of actually having the disease is only $95 / 590 = \sim 16\%$.

This is why AI algorithms must account for the *Prior Probability*, not just the accuracy of a single feature!

AI Applications of Bayes' Theorem



Spam Filtering

Classifying emails based on word occurrences.



Medical AI

Predicting diseases from symptoms & scans.

Face Recognition

Identifying individuals under noisy conditions.



Fraud Detection

Flagging unusual credit card transactions.



Weather Forecasting

Predicting climate patterns from historical data.



Speech Recognition

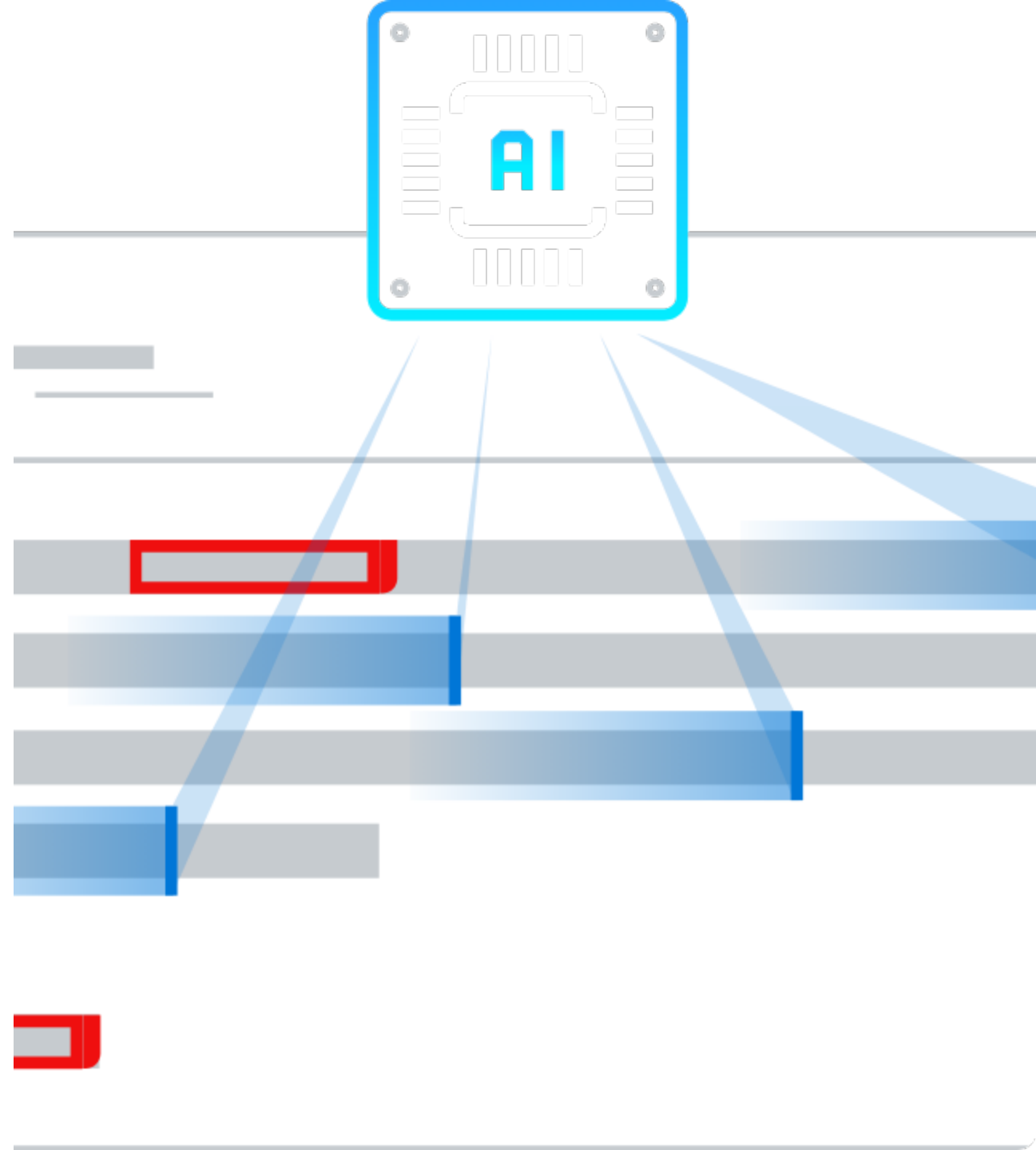
Guessing the correct word based on context.

Case Study

Spam Email Detection

A "Naive Bayes Classifier" is one of the most successful early machine learning algorithms.

- **Prior:** 20% of all emails are spam.
- **Evidence:** Email contains the word "Winner".
- **Update:** The AI mathematically updates the probability. If it crosses 90%, it redirects to the Spam folder.



Advantages & Limitations



Advantages

- ✓ Effectively handles uncertainty.
- ✓ Continuously learns from new evidence.
- ✓ Provides transparent probability scores.
- ✓ Highly scalable for text classification.



Limitations

- ✗ Requires accurate prior probabilities.
- ✗ "Naive" Bayes assumes features are independent (often false).
- ✗ Highly dependent on data quality.



Board Activity

Let's solve a simple Bayes' Theorem example together.

Prior Belief → New Evidence → Updated Belief

Common Misconceptions

-  **"90% probability means certainty."**
Correction: 90% means 1 out of 10 times, the opposite will happen. It is confidence, not a guarantee.
-  **"Bayes' Theorem predicts the exact future."**
Correction: It provides the most logical mathematical belief based strictly on past and present data.
-  **"More data always guarantees perfect predictions."**
Correction: If the data is biased or assumptions are wrong, the posterior probability will also be flawed.

Key Takeaways

- ★ **AI navigates uncertainty:** Real-world AI relies on probabilistic reasoning, not just strict logic.
- ★ **Probability = Confidence:** It acts as a mathematical measure of belief in a hypothesis.
- ★ **Belief updating:** Bayes' Theorem allows systems to systematically update beliefs as new evidence arrives.
- ★ **Ubiquitous Applications:** Powers spam filters, medical diagnostics, forecasting, and more.

Concept Check: Quick Quiz

1. Why is uncertainty important in AI?

- ▶ Because real-world data is inherently noisy and incomplete.

2. What does the "Prior Probability" represent?

- ▶ Our initial belief before receiving any new evidence.

3. What is the "Posterior Probability"?

- ▶ The updated belief calculated after analyzing the new evidence.

4. Does Bayes' Theorem completely eliminate uncertainty?

- ▶ No, it quantifies it and gives the mathematically optimal updated probability.

Coming Up Next

Unit III: Intelligent Agents

Now that we know how AI *reasons* under uncertainty, how does it physically or digitally *act* in the world?

Next lecture, we will explore:

-  What makes an agent "intelligent"?
-  Perceiving the environment via sensors.
-  The PEAS Framework.



Image Sources



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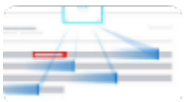
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